What can waves tell us?

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Distinguish a signal's travel from the motion of its source.

SUCCESS CRITERIA - I can...

1. Identify what a wave can carry.
2. Separate signal travel time from source speed.
3. Name evidence needed to interpret an observation.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use "wave" and "signal" to compare your model or result. Sentence starters are optional.

wave: a disturbance that can carry energy and information

signal: a disturbance that carries information

evidence: observations or measurements that support an idea

01 Bring back an earlier idea

A sphere's radius becomes 3 times as large at fixed density. Derive the area-to-mass multiplier.
Explain why kinetic energy per mass alone does not predict survival during entry.

Word help: derive = show the reasoning or algebra that produces a relationship

Sentence starter: I predict … . My model supports this because … .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: distance = speed × time
t = d/c for light in vacuum.
A signal takes time to travel. Receiving its light does not mean the source traveled with it.

Worked reasoning: A pulse travels 4 m along a rope in 2 s, so its average propagation speed is 2 m/s. A marked rope segment moves locally; it does not ride 4 m along with the pulse.

Your turn: Sketch three pulse positions and mark the same rope segment in each snapshot.

Sentence starter: The … in my model represents … . Its label shows … .

Marzano check - circle: ① ② ③

03 Represent and interpret

A pulse travels along a rope while each marked rope segment moves mainly up and down. Does the material travel all the way with the pulse?

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: ① ② ③

04 Solve and justify

A radar echo returns after .020 s. Using $c=3.0\times10^8$ m/s, find target range.
Explain the factor of 2.

Sentence starter: Under the assumption …, the evidence … supports … because … .

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

What can waves tell us?

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Analyze the reasoning

A student says seeing an asteroid’s light tells us exactly where it is “right now.” Explain one limitation.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: 1 2 3

06 Choose a method

Two objects at 290 K have different masses and materials. Must their internal energies be equal?
Give two reasons.

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: 1 2 3

07 Exit check

Two echoes return at .010 and .016 s with the same wave speed.
Compare ranges and explain whether echo delay alone gives mass.

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A signal travels 300 m in 0.9 s with timing bounds 0.85-0.95 s.
Find central speed and bounds, treating distance as exact.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3